Ps 03

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1 Problem 1 DONE

1.1 Task

Consider the curve $f : [0,1] \to \mathbb{R}^2$ given by

$$f(t) = \begin{pmatrix} t \\ \frac{2}{3}t^{3/2} \end{pmatrix}$$

Determine the length of f.

1.2 Solution

We have

$$f'(t) = \begin{pmatrix} 1 \\ t^{1/2} \end{pmatrix}$$

Both components are continuous on $[0,1]$, so f is C^1.

By Theorem 16.4.7:

$$\begin{aligned} \ell(f) &= \int_0^1 \sqrt{|f_1'(t)|^2 + |f_2'(t)|^2}\, dt \\ &= \int_0^1 \sqrt{1+t}\, dt \\ &= \left[\frac{2}{3}(1+t)^{3/2}\right]_0^1 \\ &= \frac{2}{3}\big(2\sqrt{2} - 1\big). \end{aligned}$$

2 Problem 2 DONE

2.1 Task

Write Julia code to approximate the length of a curve $f : [0,1] \to \mathbb{R}^2$ by implementing two approaches:

(a) Given a partition P of $[0,1]$, determine the polygonal length of f along P.

(b) Approximate the length integral by the mid-point rule.

(c) Test and compare your code for several illustrative curves.

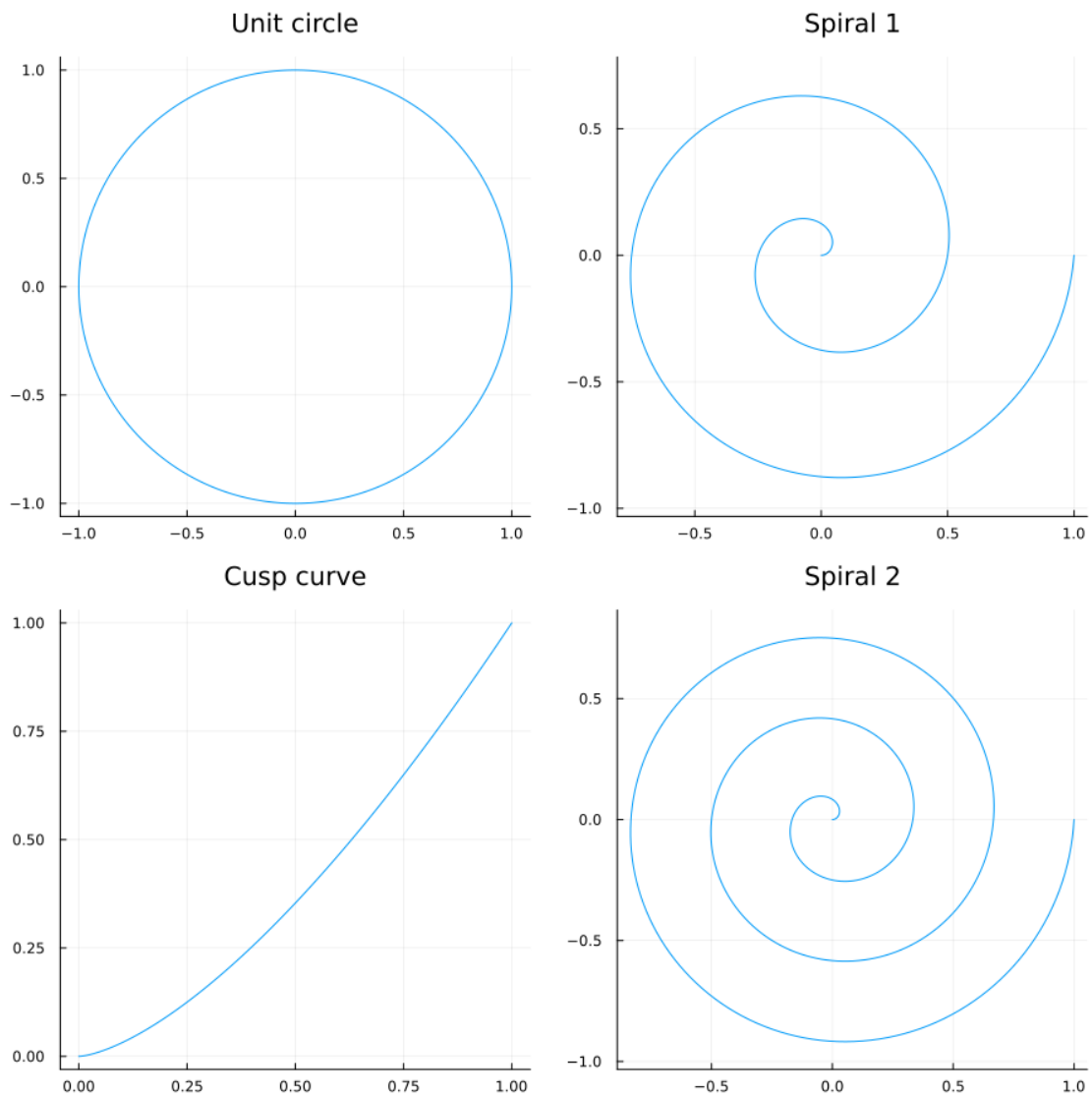
2.2 Solution

2.2.1 (a) + (b) + (c)

```
using Plots

# distance
d(x, y) = sqrt((x[1] - y[1])^2 + (x[2] - y[2])^2)

# a
function approach_a(f, P = range(0, 1, length=1000))
    l = 0.0
    for i in 1:(length(P) - 1)
        l += d(f(P[i]), f(P[i+1]))
    end
    return l
end

# b
function approach_b(f, n=1000)
    h = 1.0 / n
    s = 0.0
    for i in 0:(n-1)
        mid = (i + 0.5) * h
        v = f(mid)
        s += sqrt(v[1]^2 + v[2]^2) * h
    end
    return s
end

# curves
f_1(t) = (cos(2*t), sin(2*t))
f_2(t) = (t*cos(4*t), t*sin(4*t))
f_3(t) = (t^2, t^3)
f_4(t) = (t*cos(6*t), t*sin(6*t))

# derivatives
f_1(t) = (-2*sin(2*t), 2*cos(2*t))
f_2(t) = (cos(4*t) - 4*t*sin(4*t), sin(4*t) + 4*t*cos(4*t))
f_3(t) = (2t, 3t^2)
f_4(t) = (cos(6*t) - 6*t*sin(6*t), sin(6*t) + 6*t*cos(6*t))

function main()
    curves = [(f_1, f_1, "Unit circle"),
              (f_2, f_2, "Spiral 1"),
              (f_3, f_3, "Cusp curve"),
              (f_4, f_4, "Spiral 2")]
```

```
    # print comparison
    for (f, f, name) in curves
        a = approach_a(f)
        b = approach_b(f)
        println("$name: polygonal = $a, midpoint = $b")
    end

    # plot each curve
    ts = range(0, 1, length=500)
    p = plot(layout=(2, 2), size=(900, 900), legend=false)
    for (i, (f, _, name)) in enumerate(curves)
        xs = [f(t)[1] for t in ts]
        ys = [f(t)[2] for t in ts]
        plot!(p[i], xs, ys, title=name, aspect_ratio=:equal)
    end
    savefig("./assets/p_02.png")
end

main()
```

```
d (generic function with 1 method)
approach_a (generic function with 2 methods)
approach_b (generic function with 2 methods)
f_1 (generic function with 1 method)
f_2 (generic function with 1 method)
f_3 (generic function with 1 method)
f_4 (generic function with 1 method)
f_1 (generic function with 1 method)
f_2 (generic function with 1 method)
f_3 (generic function with 1 method)
f_4 (generic function with 1 method)
main (generic function with 1 method)
Unit circle: polygonal = 6.283174951057152, midpoint = 6.283185307179473
Spiral 1: polygonal = 6.431350516692413, midpoint = 6.43139630404216
Cusp curve: polygonal = 1.4397098361890286, midpoint = 1.4397097024672785
Spiral 2: polygonal = 9.534181365684656, midpoint = 9.534328513260087
"/Users/ziro2173/personal/university/01_bachelor/02_semester/01_analysis/02_exercises/set_03/ass
```



3 Problem 3 DONE

3.1 Task

Consider the curve $f : [0, 1] \to \mathbb{R}^2$ given by

$$f(t) = \begin{pmatrix} t\cos(6\pi t) \\ t\sin(6\pi t) \end{pmatrix}$$

(a) Determine the length of f by your best effort (how many digits are you able to determine accurately?).

(b) Write a Julia function, whose input is a curve $\gamma : [0, 1] \to \mathbb{R}^2$ and a point $t_0 \in [0, 1]$ that returns an (affine) linear function that approximates γ in a neighborhood of t_0.

(c) Visualize the affine linear approximation for f and for several other different curves.

3.2 Solution

3.2.1 (a)

Using previous code - 15 digits.

$\ell(f) = 9.53418136568465\ldots$

3.2.2 (b) + (c)

```
using Plots

# tangent line approximation
function tangent_line(, t, h=1e-7)
     = ((t + h) .- (t - h)) ./ (2h)
    p = (t)
    return t -> (p[1] + [1] * (t - t), p[2] + [2] * (t - t))
end

# curves
f_1(t) = (cos(2*t), sin(2*t))
f_2(t) = (t*cos(4*t), t*sin(4*t))
f_3(t) = (t^2, t^3)
f_4(t) = (t*cos(6*t), t*sin(6*t))

function main()
    curves = [(f_1, "Unit circle"),
              (f_2, "Spiral 1"),
              (f_3, "Cusp curve"),
              (f_4, "Spiral 2")]

    t_points = [0.2, 0.5, 0.8]
    ts = range(0, 1, length=500)
    p = plot(layout=(2, 2), size=(900, 900), legend=false)

    for (i, (, name)) in enumerate(curves)

        # plot the curve
        xs = [(t)[1] for t in ts]
        ys = [(t)[2] for t in ts]
        plot!(p[i], xs, ys, color=:blue, lw=2, title=name, aspect_ratio=:equal)

        # plot tangent lines at selected points
```

```
        for t in t_points
            pt = (t)
            scatter!(p[i], [pt[1]], [pt[2]], color=:red, ms=5)
            L = tangent_line(, t)
             = 0.05
            t_local = range(t - , t + , length=50)
            lx = [L(t)[1] for t in t_local]
            ly = [L(t)[2] for t in t_local]
            plot!(p[i], lx, ly, color=:red, lw=1.5, linestyle=:dash)
        end
    end

    savefig("./assets/p_03.png")
end

main()
```

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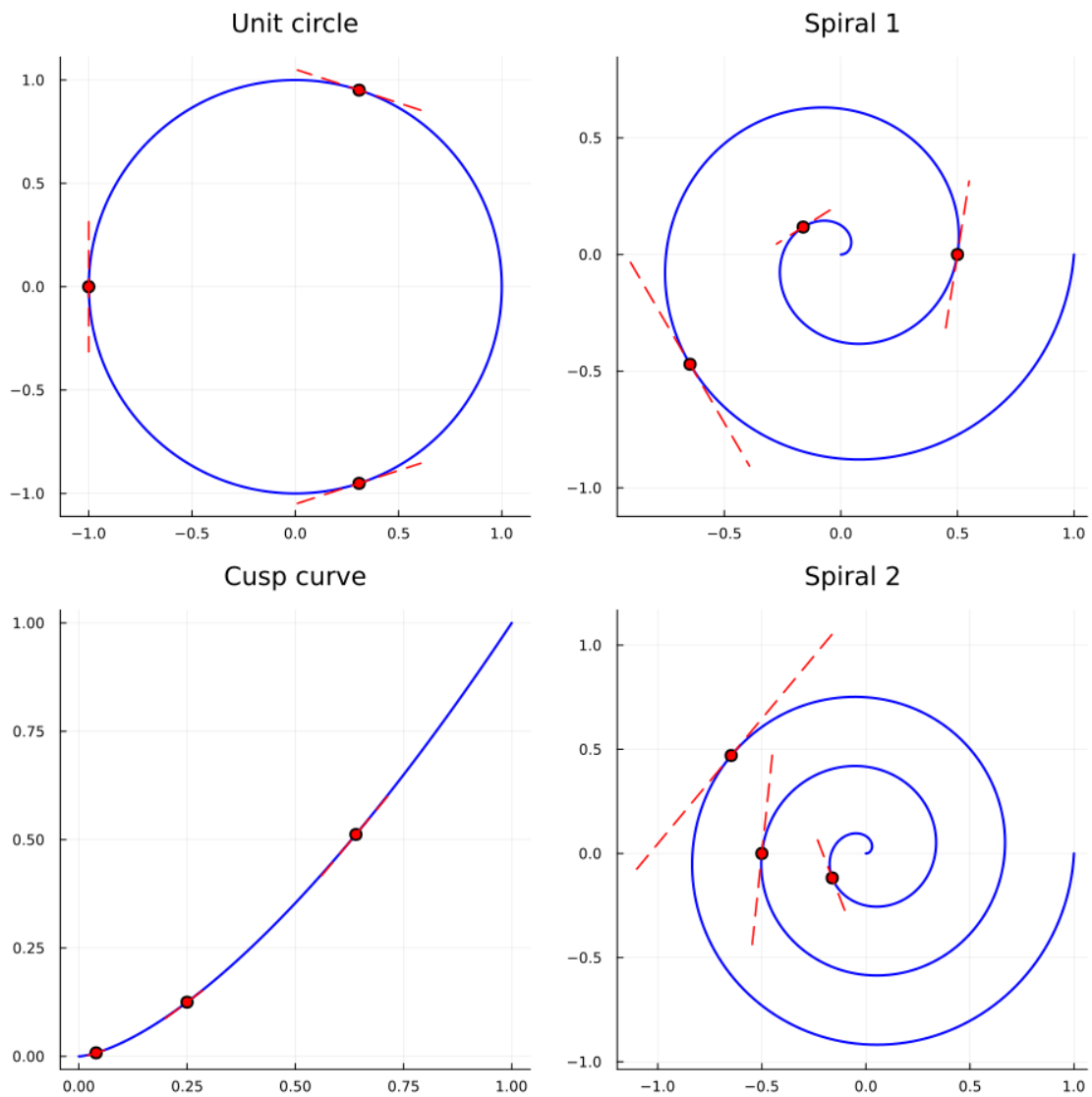

4 Problem 4 DONE

4.1 Task

Consider the upper semicircle

$$f : [0, \pi] \to \mathbb{R}^2, \qquad f(x) = \begin{pmatrix} \cos x \\ \sin x \end{pmatrix}$$

and $\varphi : [0, 1] \to [0, \pi]$ given by $\varphi(t) = \pi t^2$. Let

$$\gamma := f \circ \varphi$$

(a) Determine the metric derivative of γ.

(b) Compute the length of γ.

4.2 Solution

4.2.1 (a)

$$\gamma(t) = \begin{pmatrix} \cos(\pi t^2) \\ \sin(\pi t^2) \end{pmatrix}$$
$$\gamma'(t) = \begin{pmatrix} -\sin(\pi t^2) \cdot 2\pi t \\ \cos(\pi t^2) \cdot 2\pi t \end{pmatrix}$$

4.2.2 (b)

The length is the integral of the metric derivative:

$$\ell(\gamma) = \int_0^1 \|\gamma'(t)\| \, dt$$

Computing the norm:

$$\begin{aligned} \|\gamma'(t)\| &= \sqrt{(-\sin(\pi t^2) \cdot 2\pi t)^2 + (\cos(\pi t^2) \cdot 2\pi t)^2} \\ &= \sqrt{4\pi^2 t^2 \sin^2(\pi t^2) + 4\pi^2 t^2 \cos^2(\pi t^2)} \\ &= \sqrt{4\pi^2 t^2 \left(\sin^2(\pi t^2) + \cos^2(\pi t^2)\right)} \\ &= \sqrt{4\pi^2 t^2} = 2\pi t \end{aligned}$$

since $t \geq 0$ on $[0, 1]$. So the length is:

$$\ell(\gamma) = \int_0^1 2\pi t \, dt = 2\pi \cdot \frac{t^2}{2}\Big|_0^1 = \pi$$

This is expected: γ is a reparametrization of the upper semicircle f, which has length π. Reparametrization preserves length.